MATH 101

Homework 4

Linear and quadratic functions

Sections 4.1 and 5.1 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Section 4.1 - Linear functions

1. For each table, decide whether a linear function fits all points. Justify using rates of change. If linear, find its rule.

x	0	1	2	3
A(x)	2	5	8	11
B(x)	2	4	8	16

2. State the slope and y-intercept; decide whether the function increases or decreases. Find f(-2) and solve f(x) = 1.

$$f(x) = -\frac{3}{2}x + 4$$

3. Find the equation of the line through the two points. Use function notation.

$$(-2, 5), \quad (4, -1)$$

4. Find equations of the lines through (2, -1) that are (a) parallel and (b) perpendicular to the given line.

$$y = \frac{2}{3}x + 4$$

5. A tank contains 180 liters and loses 6 liters per minute at a constant rate. Write V(t), interpret its slope and intercept with units, and give a physically meaningful domain.

Section 5.1 - Quadratic functions

6. Write in vertex form. Give the vertex, intercepts, and increasing/decreasing intervals; sketch the parabola.

$$f(x) = x^2 - 4x - 5$$

7. Describe the transformations from y = x squared. Give the vertex, axis of symmetry, domain, range, intercepts, and increasing/decreasing intervals. Sketch the graph.

$$g(x) = -2(x + 1)^2 + 8$$

8. Find a quadratic function with vertex (3, -4) whose graph passes through (1, 4). Write it in vertex form and standard form; find both x-intercepts.

9. A rectangle has perimeter 40 meters. Let x be its width. Write its area A(x) as a quadratic function and state the physically meaningful domain. Find the dimensions giving the maximum area and state that area. Also find the dimensions giving area 75 square meters.